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JEE ADVANCED REVISION PART TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



SEC:OSR_*CO-SC(MODEL-A&B)

Time: 3 Hrs

Date: 04-04-24

Max. Marks: 180

04-04-24_OSR.IIT_STAR CO-SC(MODEL-A&B) _JEE ADV_RPTA-3_SYLLABUS

MATHEMATICS: Indefinite Integration, Definite Integration, Areas, Differential Equations

PHYSICS: Magnetism, EMI, AC, EM Waves

CHEMISTRY: Hydrocarbon, Benzene, Alkyl halide, Aryl halide & Grignard reagent, Alcohol

TIME: 3HRS

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 9 – 14)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 15 – 18)	Single Correct option type	+3	-1	4	12
Total				18	60

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 26)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 27 – 32)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 33 – 36)	Single Correct option type	+3	-1	4	12
Total				18	60

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 44)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 45 – 50)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 51 – 54)	Single Correct option type	+3	-1	4	12
Total				18	60

THE NARAYANA GROUP

MATHEMATICS**MAX.MARKS: 60****SECTION – 1 (Maximum Marks: 24)**

This section contains SIX (08) questions.

The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric

keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)**Negative Marks:** -1 In all other cases.

1. The value of $\sum_{r=0}^n \frac{r+1}{r+2} \frac{n!}{(r-1)!(n-r)!} \int_0^x t^{r-1} dt$ equal to
- A) $\frac{(1+x)^{n+2} - (n+2)x - 1}{(n+1)(n+2)x^2} - \frac{(1+x)^{n+1} - 1}{(n+1)x} + (1+x)^n$
- B) $\frac{(1+x)^{n+1} - (n+2)x - 1}{(n+1)(n+2)x^2} - \frac{(1+x)^n - 1}{(n+1)x} + (1+x)^n$
- C) $\frac{(1+x)^{n+2} - (n+1)x - 1}{(n+1)(n+2)x^2} - \frac{(1+x)^{n+1} - 1}{(n+1)x} + (1+x)^n$
- D) $\frac{(1+x)^{n+2} - (n+2)x - 1}{(n+1)(n+2)x^2} - \frac{(1+x)^{n+1} - 1}{(n+1)x} + \frac{(1+x)^n}{x}$
2. Suppose f is a real valued differentiable function defined on $[1, \infty)$ with $f(1) = 1$. Suppose $f'(x) = \frac{x^2 + 1}{x^4 + (f(x))^2}, \forall x \geq 1$, then
- A) minimum value of $f(x)$ in $[1, \infty)$, is 1 B) maximum value of $f(x)$ in $[1, \infty)$, is 1
- C) $f(x) < 1 + \frac{\pi}{2\sqrt{2}}, \forall x \geq 1$ D) $f(x) > 1 + \frac{\pi}{2\sqrt{2}}, \forall x \geq 1$
3. Let $f(x) = x + \frac{x^3}{1.3} + \frac{x^5}{1.3.5} + \frac{x^7}{1.3.5.7} + \dots \infty$
- $g(x) = 1 + \frac{x^2}{2} + \frac{x^4}{2.4} + \frac{x^6}{2.4.6} + \dots \infty$
- A) $\int_0^x e^{\frac{-t^2}{2}} dt = \frac{f(x)}{g(x)}$ B) $f'(x) = x f(x) + 1$
- C) $\int_1^\infty \frac{1}{(g(x))^2} \leq \frac{1}{2e}$ D) $f(x) + g(x) = e^x$

4. A real valued function $f(x)$ is such that $f: \left(0, \frac{\pi}{2}\right) \rightarrow \mathbb{R}^+$ satisfies the differential equation $x f'(x) = 1 + f(x) \{x^2 (f(x)) - 1\}$ and $f\left(\frac{\pi}{4}\right) = \frac{4}{\pi}$, then
- A) $f(x) = \frac{\sin x}{x}$ B) $f(x) = \frac{\tan x}{x}$ C) $\lim_{x \rightarrow 0} f(x) = 1$ D) $f(x) = \frac{\tan^{-1} x}{x}$
5. If $\frac{dy}{dx} = \frac{(\sin y) + x}{\sin 2y - x \cos y}$ and $y(0) = \frac{\pi}{2}$, then
- A) If $y(a) = \frac{-\pi}{2}$ then $a = 0, 2$ B) If $y(a) = \frac{-\pi}{2}$ then $a = 1, 2$
- C) $\sin y \in \left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right)$ D) $\sin y \in \left[-1, -\sqrt{\frac{2}{3}}\right] \cup \left[\sqrt{\frac{2}{3}}, 1\right]$
6. Area bounded by $y = f^{-1}(x)$ and tangent and normal drawn to it at the points with abscissa π and 2π respectively, where $f(x) = \sin x - x$ is $\frac{\pi^2}{a} - b$, here $a, b \in \mathbb{N}$, then which of the following is/are correct?
- A) a, b are co primes B) $a + b$ is even integer
- C) $a + b$ is odd integer D) a is prime number

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

9. If $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n k \ln \left(\frac{n^2 + (k-1)^2}{n^2 + k^2} \right)$ exists and is equal to L . Then the absolute value of $[L]$ is

[Note: $[y]$ denotes greatest integer less than or equal to y]

8. Let $y = f(x)$ be a function such that

$$\int_0^k [x] f'(x) dx = g(k) f(k) - \sum_{n=1}^{[k]} f(n)$$

Number of points of discontinuity of $y = g(x)$ in the interval $(0, \pi)$ is _____

Where $[\cdot]$ denotes greatest integer function.

9. The value of $\lim_{n \rightarrow \infty} \frac{1}{e^{n-1}} \int_0^n \frac{dx}{e^x (e^{x+3} + 2e^2 - 1)}$ equal to

10. Let $f(x)$ be a non-constant function such that $\int_0^x (f(t))^3 dt = \frac{1}{x^2} \left(\int_0^x f(t) dt \right)^3 \quad \forall x \in \mathbb{R} - \{0\}$.

If $f(1) = 1$ then $f\left(\frac{1}{2}\right) =$ _____

11. Let $f(x)$ be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that

$$\lim_{t \rightarrow x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1 \text{ then } \lim_{x \rightarrow 1} \frac{\log_e^{f(x)} - \log_e^2}{x-1} =$$

12. Let $\lim_{n \rightarrow \infty} \left[{}^nC_0 {}^nC_1 {}^nC_2 \dots {}^nC_n \right]^{\frac{1}{n(n+1)}} = L$. Find the value of $\log_e(L^2)$.

13. $f(-1, 1) \rightarrow \mathbb{R}$ be a differentiable function satisfying

$f'(x) = 2\sqrt{f(x)}, \forall x \in (-1, 1)$ and $f(0) = 0$. If number of such function satisfying given

conditions is 'a'. Find $\lim_{x \rightarrow a} \frac{x^2 + 4}{x^2 + 2}$.

14. Let A_r be the area of region bounded between the curves $y^2 = (e^{-kr})x, k > 0, r \in \mathbb{N}$ and the line $y = mx, m > 0, k$ and m are some constants then if $\lim_{n \rightarrow \infty} \sum_{r=1}^n A_r = \frac{4}{3(e^{2k} - 1)}$, then

$6m =$ _____

SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

15. Match the following List-I with List-II

	List-I		List-II
(P)	If function f satisfies $\int_0^1 (f^4(x) - 8x^2 \cdot f^2(x)) dx = -\frac{16}{5}$, then the value of $f^2(1)$ is	(1)	0
(Q)	If the point of inflection for the curve $y = \int_0^x \sqrt{1-t^4} dt$ is (α, β) , then $\alpha + \beta$ is equal to	(2)	1
(R)	Let f be continuous and function g is defined as $g(x) = \int_0^x \left(t^2 \cdot \int_1^x f(u) du \right) dt$ also $f(1) = 3, g'(1) + g''(1)$ is equal to $n_1 + \frac{f'(1)}{n_2}$ (where $n_1, n_2 \in \mathbb{N}$), then $n_1 - n_2$ is equal to	(3)	2
(S)	Let $\ell_1 = \int_0^1 \frac{e^x}{1+x} dx$ and $\ell_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)} dx$, then $\frac{\ell_1}{e\ell_2}$ is	(4)	3
		(5)	4

A) P-4;Q-1;R-3;S-4

B) P-5;Q-1;R-2;S-3

C) P-1;Q-2;R-3;S-4

D) P-5;Q-1;R-5;S-4

16. Match the following.

	List-I		List-II
(P)	Area bounded by the curve $y = \sin^{-1} x = x - x^2$, and the x-axis is equal to	(1)	0
(Q)	The area bounded by $y = x^2 - 3$ and the line $y = ax + 2$ attains its minimum value than the parameter 'a' is equal to	(2)	1
(R)	The area of the region represented by the expression $\sqrt{2} \leq x+y + x-y \leq 2\sqrt{2}$	(3)	$\pi - 1$
(S)	If k is a positive number and the area of the region bounded by the curves $y = x - kx^2$ and $ky = x^2$ attains its maximum value, then k is equal to	(4)	6

The correct option is

A) P-3;Q-4;R-2;S-1

B) P-1;Q-4;R-2;S-3

C) P-4;Q-4;R-3;S-2

D) P-3;Q-1;R-4;S-2

17. Match the following.

	List -I		List-II
(P)	Area bounded by $y = x^2 + 2$ and $y = 2 x - \cos \pi x$ is equal to A, then 3A equals	(1)	4
(Q)	The value of $2 \cot(\cot^{-1} 3 + \cot^{-1} 7 + \cot^{-1} 13 + \cot^{-1} 21)$ is equal to	(2)	$\frac{8}{3}$
(R)	Tangent and normal at the ends A and C focal chord AC of parabola $y^2 = 4x$ intersect at B and D, then minimum area of quadrilateral ABCD is	(3)	3
(S)	Number of ordered pair (α, β) for which the area common to $x^2 + y^2 - 2\alpha^2 x - 2\beta^2 y + c = 0$ and its image in $x + y = 1$ is maximum is $(\alpha, \beta \in \mathbb{Z})$	(4)	8

A) P-4;Q-3;R-2;S-3

B) P-1;Q-2;R-1;S-3

C) P-4;Q-3;R-4;S-1

D) P-1;Q-3;R-4;S-2

18. Match the following.

	List -I		List-II
(P)	If $\int \frac{2^x}{\sqrt{1-4^x}} dx = k \sin^{-1}(f(x)) + C$, then k is greater than	(1)	0
(Q)	If $\int \frac{(\sqrt{x})^5}{(\sqrt{x})^7 + x^6} dx = a \ln \frac{x^k}{x^k + 1} + c$, then ak is less than	(2)	1
(R)	If $\int \frac{x^4 + 1}{x(x^2 + 1)^2} dx = k \ln x + \frac{m}{1+x^2} + n$, where n is the constant of integration, then mk is greater than	(3)	3
(S)	If $\int \frac{dx}{5 + 4 \cos x} = k \tan^{-1}\left(m \tan \frac{x}{2}\right) + C$, then k/m is greater than	(4)	4

A) P-1,2;Q-1,4;;R-1,4;S-3

B) P-1,2;Q-2,4;R-3;S-2,3

C) P-1,2;Q-3,4;R-1;S-1,2

D) P-1,2;Q-3,4;R-1,4;S-1,2

PHYSICS

MAX.MARKS: 60

SECTION – 1 (Maximum Marks: 24)

This section contains SIX (08) questions.

The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9**, **BOTH INCLUSIVE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

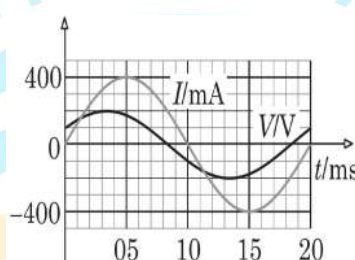
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

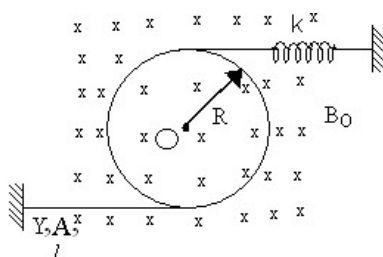
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

19. The given graph shows variation with time in the source voltage V and steady state current I drawn by a series RLC circuit. Which of the following statements is/are correct?



- A) Current lags the voltage.
 B) Resistance in the circuit is $250\sqrt{3}\Omega$.
 C) If capacitive reactance is 74Ω , inductance in the circuit is approximately 560 mH.
 D) Average power dissipation in the circuit is $20\sqrt{3}W$.
20. A uniform non-conducting circular disc of mass M and radius R charged uniformly with charge Q is free to rotate in horizontal plane about a vertical axis passing through its fixed centre O . A mass less spring of constant k and an elastic cord of length l , cross-sectional area A and young's modulus Y are attached to diametrically opposite points of the disc such that their lengths are parallel as shown in the figure. Initially the spring is unstretched, the cord is just taut, and a uniform magnetic field B_0 exists in vertical direction. Now B_0 is switched off. Assuming B_0 is small enough for the oscillations to be small and simple harmonic, choose the correct options



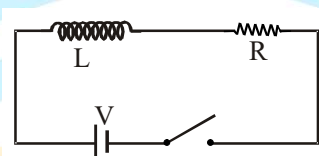
A) The time period of oscillations is $T = 2\pi \sqrt{\frac{Ml}{2(Kl + YA)}}$

B) The time period of oscillations is $T = \pi \sqrt{\frac{Ml}{2(Kl + YA)}} + \pi \sqrt{\frac{M}{2K}}$

C) During the oscillations the maximum potential energy stored in the spring is $\frac{Q^2 B_o^2 R^2}{16M}$

D) During the oscillations the maximum elastic potential energy stored in the cord is $\frac{Q^2 B_o^2 R^2}{16M}$

21. A resistor and an inductor in series are connected to a battery through a switch. After the switch is closed (at $t = 0$)



A) Joule heat dissipation in the resistor changes at fastest rate at $t = \frac{L}{R} (\ln(2))$

B) When the rate of increase of magnetic energy stored in the coil is at maximum, current in the circuit is $\left(\frac{V}{2R}\right)$

C) Power dissipated in resistor increases continuously but rate of change of power initially increases, it later reaches a maximum, and beyond a point decreases to zero

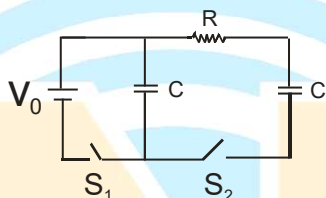
D) The fastest increase of current takes place at $t = \left(\frac{V}{2R}\right)$.

22. A parallel plate capacitor of capacitance C_0 is charged to a voltage V and then the battery is disconnected. A dielectric covering one-third area of each plate is now inserted as shown in the figure. If charges on the capacitor plates get redistributed such that the portions covered with dielectric and not covered with the dielectric share equal amounts of charge, which of the following statements is/are true?

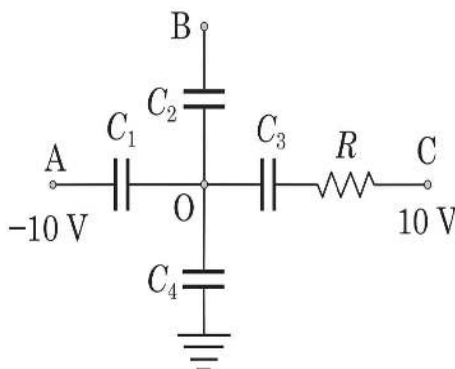


- A) Dielectric constant of the dielectric is 2.0.
- B) Charge appearing due to polarization on the surface of the dielectric is $0.25 C_0 V$.
- C) Force of electrostatic interaction between portions of the plates covered with dielectric is equal to that between uncovered portions.
- D) Force of electrostatic interaction between the plates after insertion of the dielectric becomes $9/8$ times of its value before insertion of the dielectric

23. In a circuit shown in figure, the capacitance of each capacitor is equal to C and the resistance equal to R . One of the capacitance was connected to a voltage V_0 by closing switch S_1 . Now at $t = 0$, the switch S_1 is opened and S_2 is closed



- A) The current in the circuit as a function of time t is $I = \frac{V_0}{R} e^{-2t/RC}$
- B) Charge as a function of time is $q = \frac{CV_0}{2} (1 - e^{-2t/RC})$
- C) Heat produced in the circuit is $\frac{CV_0^2}{2}$
- D) Heat produced in the circuit is $\frac{CV_0^2}{4}$
24. The figure shows part of large network in which potentials of some of the points are shown. Each capacitor has a capacitance $5 \mu F$. Which of the following statements is/are true?



- A) From the given information, potential of point O can be determined but that of B cannot be determined.
- B) From the given information, potential of both the points O and B cannot be determined.
- C) If charge on capacitor C_2 were also specified, potential of point O can be determined but that of B cannot be determined.
- D) If charge on capacitor C_2 were also specified, potentials of both the points O and B can be determined

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

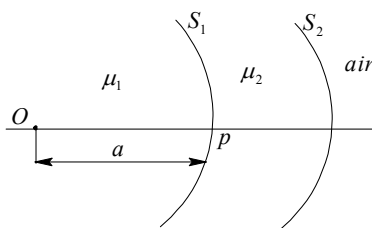
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

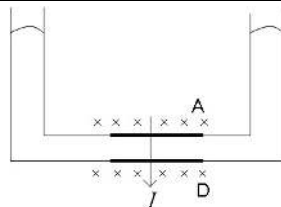
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

25. A point object 'O' is placed in a medium of refractive index $\mu_1 = 1.4$. S_1 and S_2 are two concentric spherical surfaces of radii 1m and 2m. To the right of 'O' contains a medium of refractive index $\mu_2 = 1.5$ between the interfaces S_1 and S_2 . Find the object distance of O from S_1 (in meter) so that an image of 'O' as seen by observer from air, coincides with O.

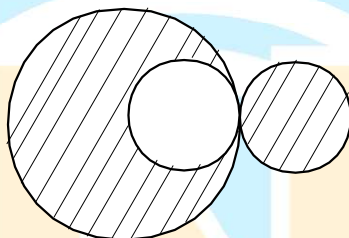


26. A conducting liquid of density $13.6 \times 10^3 \text{ kgm}^{-3}$ is contained in a u-tube of uniform square cross-sectional side $a = 4 \text{ mm}$. Electrodes A and D are sealed inside as in fig.



An electric current $I = 10\text{A}$ passes between the electrodes A to D and a magnetic field $B = 326.4 \times 10^{-3}$ tesla is applied across the horizontal arm perpendicular to the plane of the fig. and inside to the plane. Calculate the difference in liquid levels in mm in the u-tube. Surface tension is negligible. $[g = 10\text{ms}^{-2}]$

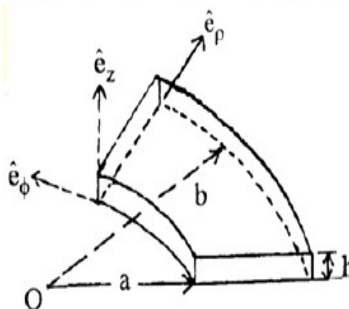
27. Consider a uniform charged sphere of charge density ρ and radius R . If the cavity of radius $R/2$ is made inside the sphere and the removed part is again placed near the charge distribution as shown in the figure. If the electric field at the centre of the cavity is $\frac{\rho R}{n\epsilon_0}$, find n .



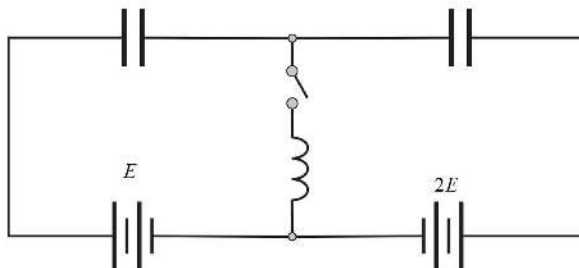
28. In a series LCR circuit, half power frequencies (frequency at which power dissipated is half the maximum power dissipated in the circuit) are found to be 90 rad/s and 40 rad/s . Find the value of $\frac{\omega_0}{10}$, where ω_0 is the resonance frequency.

29. A conducting medium is the form of a quarter of an annulus of radii a and b ($b > a$) with uniform thickness h as shown in figure. The conductivity σ of the material of the medium is uniform. The resistance of the specimen between the pair of opposite faces

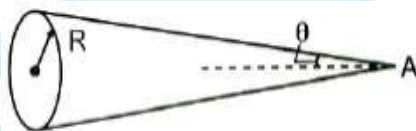
in the direction $\hat{e}_\rho$ is $\frac{X \ln\left(\frac{b}{a}\right)}{\pi \sigma h}$. Then X is



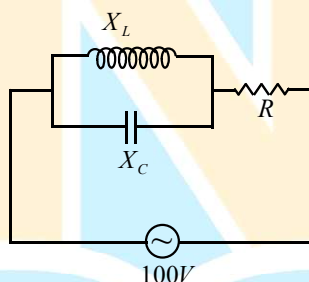
30. A circuit shown consists of two identical capacitors each of capacitance C , a coil of inductance L , two ideal batteries of electromotive force E and $2E$ and a switch. The switch is open for a long time. Find maximum current through the coil after the switch is closed is $3E\sqrt{\frac{C}{xL}}$, where $x = ?$



31. A cone made of insulating material has a total charge $Q = 3 \text{ mC}$ spread uniformly over its slopping surface. If the energy (in joule) required to take a point charge $q = 5 \mu\text{C}$ from infinity to apex A of the cone is $9z$, find z . (Given: half cone angle $\theta = 37^\circ$, $R = 2\text{m}$)



32. In the circuit shown in the figure, $X_L = 6\Omega$, $X_C = 3\Omega$ and $R = 8\Omega$. Find the current through the voltage source.

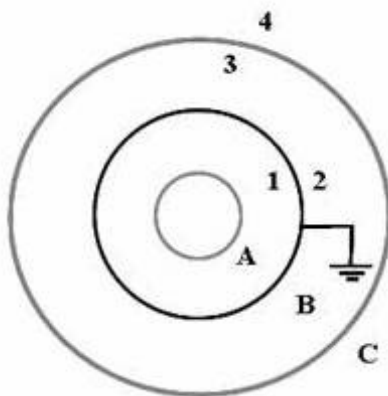


SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

33. Fig. shows three concentric thin spherical conducting shells A, B and C of radii R , $2R$ and $3R$. The shell B is earthed and A and C are given charges q and $2q$, respectively. If the charge appearing on surface 1, 2, 3 and 4 are q_1, q_2, q_3 , and q_4 , respectively, then match the following columns.



Column – I		Column – II	
A)	q_1	p)	$\frac{2}{3}q$
B)	q_2	q)	$\frac{4}{3}q$
C)	q_3	r)	$-\frac{4}{3}q$
D)	q_4	s)	$-q$

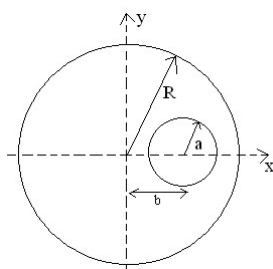
A) $A \rightarrow s; B \rightarrow r; C \rightarrow q; D \rightarrow p$

B) $A \rightarrow p; B \rightarrow q; C \rightarrow s; D \rightarrow r$

C) $A \rightarrow r; B \rightarrow p; C \rightarrow s; D \rightarrow q$

D) $A \rightarrow s; B \rightarrow q; C \rightarrow r; D \rightarrow p$

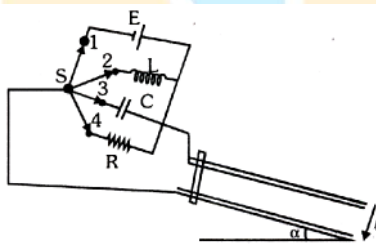
34. A very long straight conductor has a circular cross-section of radius R . Inside the conductor there is a cylindrical hole of radius ' a ' whose axis is parallel to the axis of the conductor and at a distance ' b ' from it, as shown in the diagram. The conductor with hole carries current i . Let the z -axis be the axis of conductor and the axis of the hole at $x = b$, Match the following column.



Column – I		Column – II	
A)	Magnetic field at point P(2R,0)	p)	Zero
B)	X-component of magnetic field at point P(0,0)	q)	$\frac{\mu_0 i}{2\pi b} \left(\frac{a^2}{R^2 - a^2} \right)$
C)	Magnetic field at point P(0,0)	r)	$\frac{\mu_0 i}{2\pi(R^2 - r^2)} \left(\frac{a^2}{2R - b} - \frac{R}{2} \right)$
D)	Magnetic field at point P(b,0)	s)	$\frac{\mu_0 i b}{2\pi(R^2 - a^2)}$

A) $A \rightarrow p; B \rightarrow r; C \rightarrow q; D \rightarrow s$ B) $A \rightarrow r; B \rightarrow q; C \rightarrow s; D \rightarrow p$ C) $A \rightarrow r; B \rightarrow p; C \rightarrow s; D \rightarrow q$ D) $A \rightarrow r; B \rightarrow p; C \rightarrow q; D \rightarrow s$

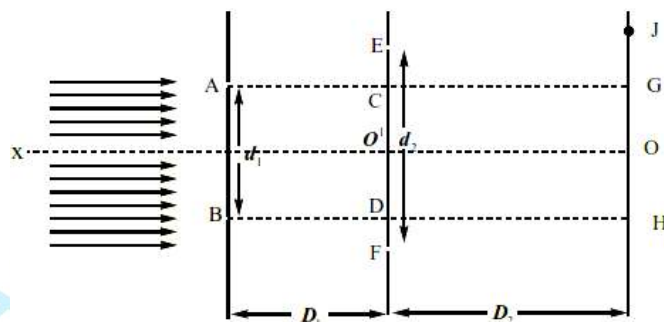
35. A homogenous field of magnetic induction B is perpendicular and into the plane of a sufficiently long track of gauge length l (separation between tracks) which is inclined at an angle α to the horizontal. A conducting rod of mass ' m ' slides on the two frictionless rails of the track as shown in the figure. The rod is released from rest. If the circuit formed by the rod and the track is closed by the moving switch of terminals 1, 2, 3 and 4. (Neglect the resistance every where except shown resistances)



	Column – I		Column – II
A)	A resistor of resistance R	P)	Rod moves with constant acceleration
B)	A capacitor of capacitance C	Q)	Rod moves with decreasing acceleration and finally attains a terminal velocity (constant)
C)	A coil of inductance L	R)	Rod will perform simple harmonic motion about uniformly accelerated mean position
D)	An ideal cell of time varying emf $E = Bl(g \sin \alpha)t$	S)	Rod moves with constant velocity

A) $A \rightarrow Q; B \rightarrow P; C \rightarrow R; D \rightarrow P$ B) $A \rightarrow R; B \rightarrow Q; C \rightarrow R; D \rightarrow P$ C) $A \rightarrow S; B \rightarrow Q; C \rightarrow R; D \rightarrow P$ D) $A \rightarrow R; B \rightarrow Q; C \rightarrow Q; D \rightarrow P$

36. In front of a screen, two parallel opaque cardboards are placed parallel to the screen. The distance between the cardboards is D_1 and the distance between the right board and the screen is D_2 . On each of these cardboards two narrow parallel slits are symmetrically cut. Distance between the slits in the first and the second screens are d_1 and d_2 respectively as shown in figure. A plane monochromatic light wave of wavelength ' λ ' is incident normally on first set of the slits



Points A, C, G are collinear and B, D, H are collinear as shown. J is a general point on screen whose Y co-ordinate with respect to point 'O' on screen is y . $OO'X$ is the X-axis and is the symmetry line.

Match the situations about intensities at different points in column I to quantity in column II.

	Column I		Column II
A)	For $\lambda = 0.2 \mu\text{m}$, $d_1 = 1 \text{ mm}$, $D_1 = 2 \text{ m}$ the ratio of intensity at C to O' is	P)	$\frac{3}{4}$
B)	For $\lambda = 0.2 \mu\text{m}$, $d_1 = 1 \text{ mm}$, $D_1 = 2 \text{ m}$, $d_2 = 40 \text{ mm}$, $D_2 = 40 \text{ m}$, $y = \frac{200}{3} \text{ mm}$, the ratio of intensity at J to O is	Q)	$\frac{1}{2}$
C)	In option B ratio of light intensity at E to O' is	R)	1
D)	In option B if initial intensity at O is I_1 then when d_1 and d_2 are reduced to $\frac{1}{50}$ th and $\frac{1}{3}$ rd of their initial values the intensity at O is I_2 . The value of $I_2 : I_1$ is	S)	$\frac{1}{4}$

A) $A \rightarrow P$; $B \rightarrow Q$; $C \rightarrow S$; $D \rightarrow P$

B) $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow R$; $D \rightarrow P$

C) $A \rightarrow P$; $B \rightarrow Q$; $C \rightarrow R$; $D \rightarrow S$

D) $A \rightarrow S$; $B \rightarrow S$; $C \rightarrow S$; $D \rightarrow S$

CHEMISTRY

MAX.MARKS: 60

SECTION – 1 (Maximum Marks: 24)

This section contains SIX (08) questions.

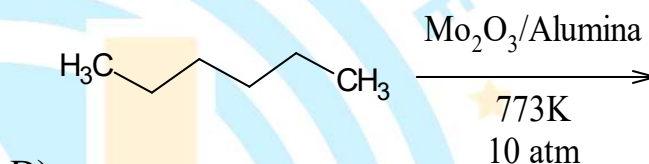
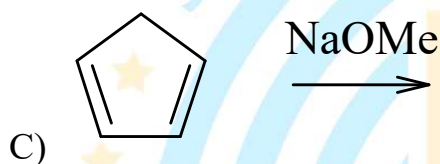
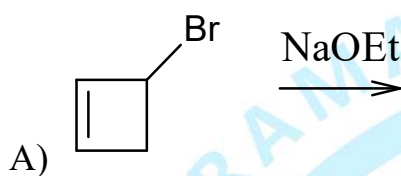
The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9**, **BOTH INCLUSIVE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

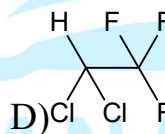
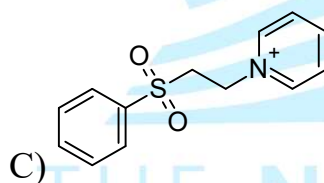
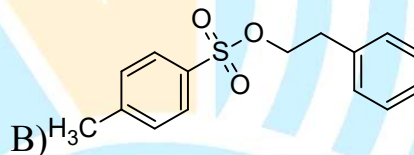
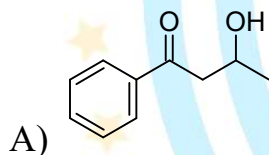
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.**Zero Marks :** 0 If none of the option is chosen.(i.e the question is un answered)**Negative Marks:** -1 In all other cases.

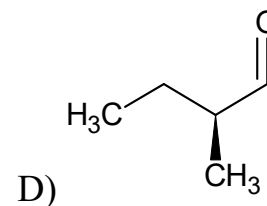
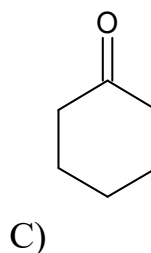
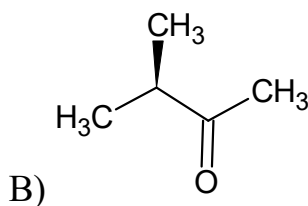
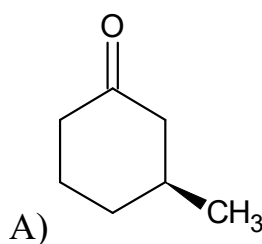
37. Choose the correct option(s) that give(s) an aromatic compound as the product (with or without other by products)



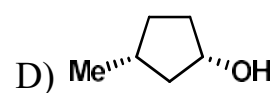
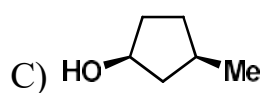
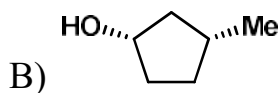
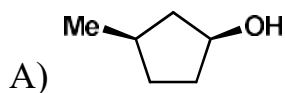
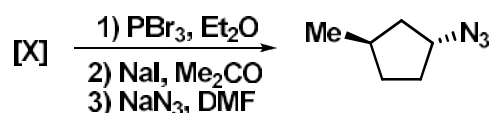
38. Which of following compound(s) are likely to undergo E^1CB path upon heating with $NaOH$?



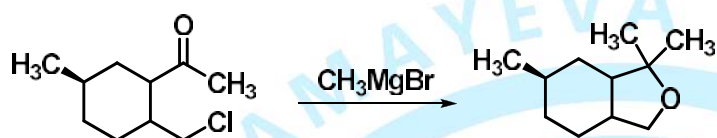
39. Which of the following give pair of diastereomers on reaction with CH_3MgBr followed by H_2O ?



40. In the following reaction sequence, the correct structure (s) of [X] is (are)



41. Correct statements regarding the given reaction are



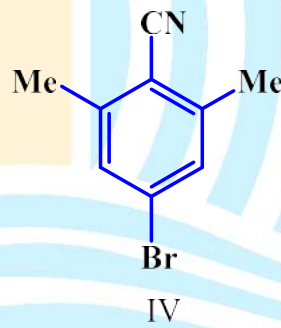
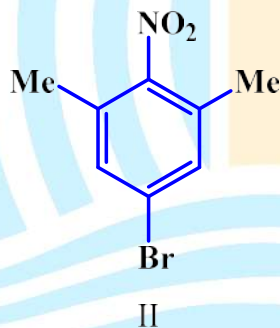
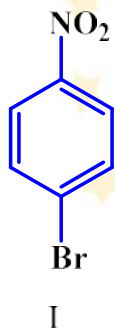
A) Nucleophilic addition

B) Nucleophilic substitution reaction

C) Carbocation is intermediate

D) Electrophilic addition

42. Select the correct statement(s) regarding the given molecules?



A) The rate difference between III and IV is more compare to I and II with NaOH

B) The rate difference between III and IV is less compare to I and II with NaOH

C) I and III react with same rate with NaOH

D) Resonance effect is more in I than II

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

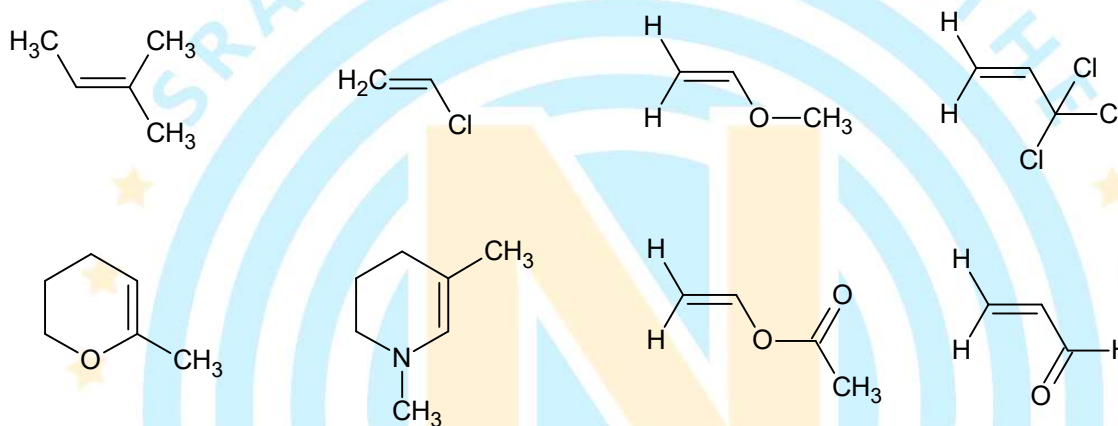
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

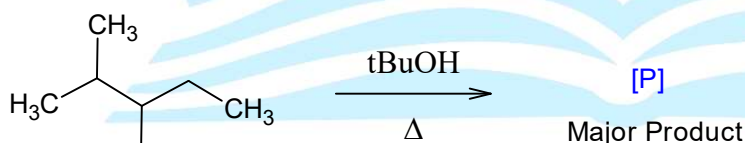
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

45. How many of the following compound(s) will give Markovnikov's product as the major product during addition of HCl?

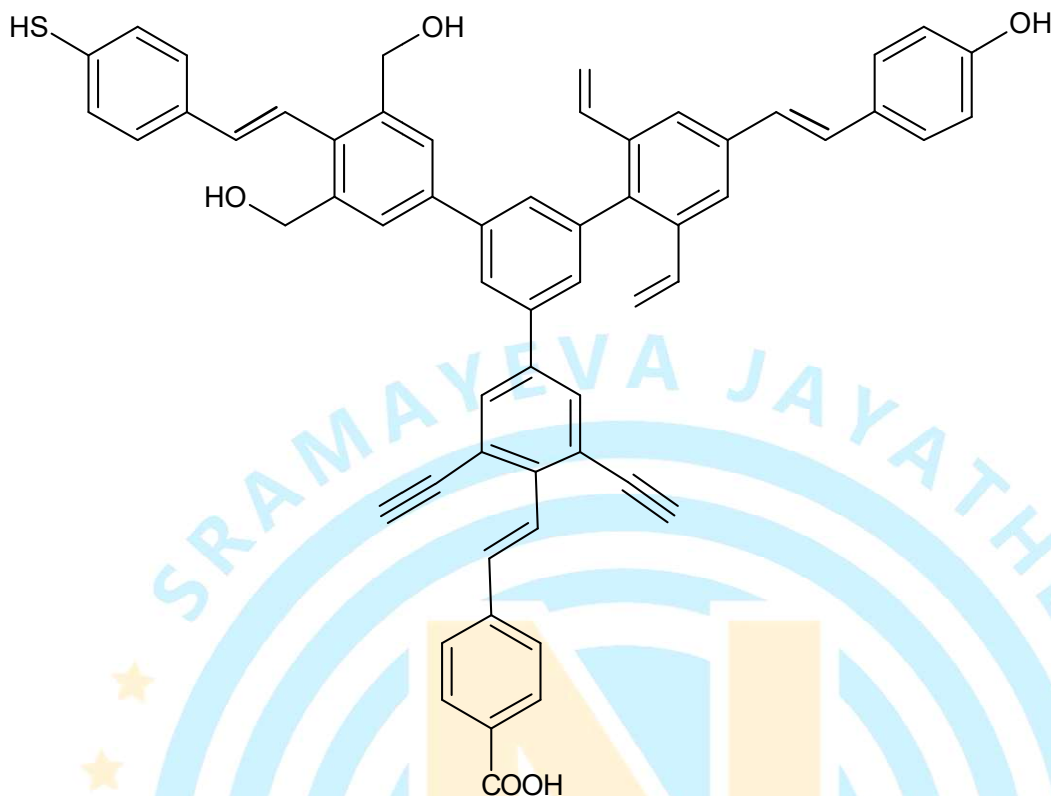


44. The number of C-H bonds that involved in hyperconjugation of major alkene [P] formed in the following reaction is -----

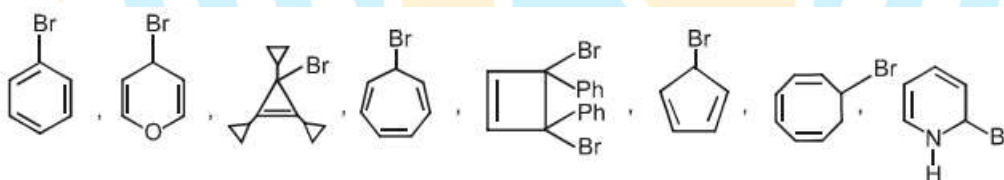


45. Zingiberene is a fragrant compound isolated from ginger, has the molecular formula $C_{15}H_{24}$ and is known not to contain any triple bond(s). When zingiberene is subjected to catalytic hydrogenation using an excess of hydrogen gas, one mole of zingiberene absorbs three moles of dihydrogen gas and produce a compound with the molecular formula $C_{15}H_{30}$. By using this data find the number of ring(s) is present in zingiberene.

46. The number of moles of CH_4 produced when 1 mole of the following compound is treated with excess methyl magnesium bromide (CH_3MgBr) is _____.



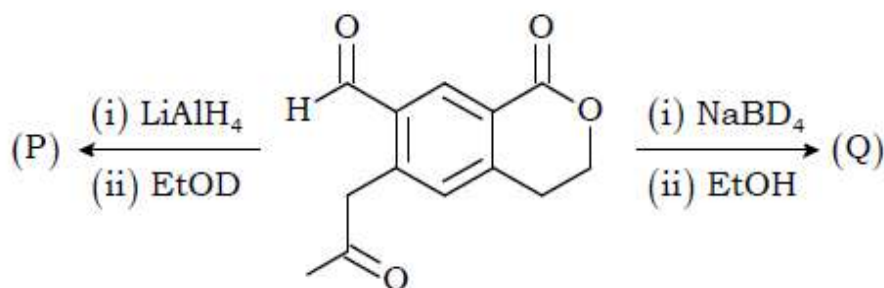
47. How many of the following will give yellow precipitate of $AgBr$, on reaction with $AgNO_3$?



48. How many resonance structures are possible for the intermediate of given reaction?



49.



(i) The number of O–D bonds in the products 'P' and 'Q' is x

(ii) The number of C–D bonds in the products 'P' and 'Q' is y

The value of $(x+y)$ is

50. PCl_3 and an alcohol can react in $x:y$ molar ratio to form corresponding alkyl halide and an oxyacid of basicity z . What is $\left(\frac{y}{x} + z\right)$?

SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

51. Match the different chemical reactions given in **column I** with respective to their characteristic properties given in **column II** and select the correct answer

	Column-I		Column-II
P)		1)	Carbocation
Q)		2)	Free radical
R)		3)	Markovnikov's addition
S)		4)	Cyclic intermediate

A) P–1, 4; Q–4; R–3; S–1

B) P–1, 3; Q–2, 3; R–1, 2; S–3

C) P–3, 4; Q–2; R–4; S–3

D) P–3, 4; Q–4; R–1, 4; S–1

52. Match the compounds in column I with the relative rate of cyclization (intermolecular S_N2 rate) in column II and select the correct answer

	Column-I		Column-II
P)		1)	0.07
Q)		2)	0.0001
R)		3)	1
S)		4)	100

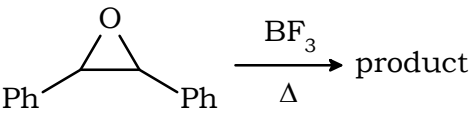
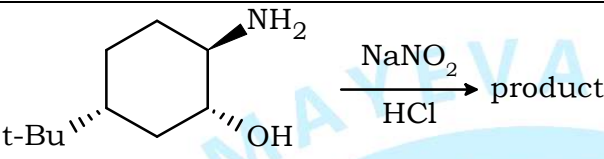
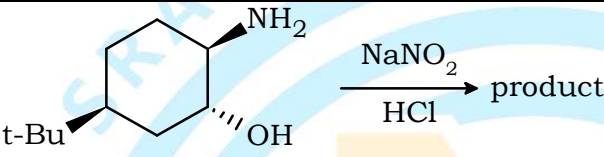
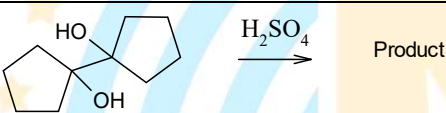
A) $P-2, Q-1, R-3, S-4$ B) $P-1, Q-2, R-3, S-4$ C) $P-2, Q-1, R-4, S-3$ D) $P-1, Q-2, R-4, S-3$

53. Match the conversion in Column-I with appropriate reagent(s) of column-II and select the correct answer

	Column-I		Column-II
P)		1)	PCC
Q)		2)	Cu/573K
R)		3)	K ₂ Cr ₂ O ₇ + dil. H ₂ SO ₄
S)		4)	H ₃ PO ₄ /358K

A) $P-2; Q-1,4; R-2,4; S-1,4$ B) $P-3; Q-1,2; R-1,2,3; S-2,4$ C) $P-1,2,3; Q-1,2,4; R-2,4; S-1,2$ D) $P-1; Q-1,2; R-1,3,4; S-2,4$

54. Match the different chemical reactions given in **column I** with respective to their characteristic properties given in **column II** and select the correct answer

	Column-I		Column-II (Take only major path)
P)		1)	Ketone
Q)		2)	Aldehyde
R)		3)	Ring contraction
S)		4)	Epoxide

A) P-3; Q-1,4; R-2; S-3

B) P-2; Q-1,4; R-2,3; S-1

C) P-3; Q-1,3; R-2,3; S-3

D) P-2; Q-2,3; R-4; S-1

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